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Chaos, Solitons and Fractals

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Dear Editors,

We hereby submit the manuscript entitled *"Spatiotemporal Dynamics of Conway's Game of Life in the Entropy–Complexity Plane"* for consideration for publication in *Chaos, Solitons & Fractals*. The work fits the journal's scope, addressing cellular automata, chaotic dynamics, complexity measures, and convergence analysis in the setting of Conway's Game of Life (GoL).

In this paper, we propose a methodological framework to characterize the structural information of GoL through the entropy–complexity ($H \times C$) plane. At each generation, the two-dimensional lattice is traversed with a Peano–Hilbert space-filling curve, which preserves spatial locality. The resulting bit sequences are converted into integer values, yielding a well-defined time series for that generation. Each series is then mapped to a point in the $H \times C$ plane through the Bandt–Pompe ordinal-pattern methodology, using Shannon permutation entropy and Jensen–Shannon statistical complexity. A simulation of T generations thus becomes a trajectory in the plane, enabling the analysis of both transient and steady-state behavior.

Compared to previous work, we explore substantially larger configurations, with lattices of 1024×1024 cells and simulations of up to 10 000 generations. Our results show that, although GoL is highly sensitive to the initial density of live cells, trajectories consistently converge to a stable region (H^*, C^*) of the plane. The convergence follows an approximately exponential decay in the early stages before saturating into a plateau, whose onset and apparent rate depend on the initial density. These findings indicate that the entropy–complexity plane, combined with the proposed Peano–Hilbert and bit-depth encoding, offers a robust and scalable framework for characterizing the dynamical regimes of cellular automata.

The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All authors have approved the manuscript and agree with its submission to *Chaos, Solitons & Fractals*. We believe the proposed methodology and findings will be of interest to the journal's readership, particularly researchers working on cellular automata, complex systems, and information-theoretic analysis of nonlinear dynamics.

Thank you for your time and consideration.

We look forward to your response.

Sincerely,

Lucas Malacarne Astore (corresponding author),
Alejandro C. Frery,
Petrônio Cândido de Lima e Silva,
Frederico Gadelha Guimarães